

Granular self-organization by autotuning of friction

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A monolayer of granular spheres in a cylindrical vial, driven continuously by an orbital shaker and subjected to a symmetric confining centrifugal potential, self-organizes to form a distinctively asymmetric structure which occupies only the rear half-space. It is marked by a sharp leading edge at the potential minimum and a curved rear. The area of the structure obeys a power-law scaling with the number of spheres. Imaging shows that the regulation of motion of individual spheres occurs via toggling between two types of motion, namely, rolling and sliding. A low density of weakly frictional rollers congregates near the sharp leading edge whereas a denser rear comprises highly frictional sliders. Experiments further suggest that because the rolling and sliding friction coefficients differ substantially, the spheres acquire a local time-averaged coefficient of friction within a large range of intermediate values in the system. The various sets of spatial and temporal configurations of the rollers and sliders constitute the internal states of the system. Experiments demonstrate and simulations confirm that the global features of the structure are maintained robustly by autotuning of friction through these internal states, providing a previously unidentified route to self-organization of a many-body system.

self-organization | granular | friction | rolling

Driven, dissipative, and nonlinear many-body systems are capable of self-organizing (1). Examples include avalanches in sand piles (2), earthquakes (3), pinned or depinned disordered systems (4), and animals in collective locomotion patterns (5). The self-organized state in these systems is usually characterized by a macroscopically observable or measurable quantity, for example, the slope of a sand pile or the appearance of large-scale coherent motion of motile objects. The robustness of these self-organized states against external perturbation is achieved through feedback processes which suitably change the internal states of the system. It is thus important to identify the relevant internal states of the system and the feedback mechanism at play to understand the self-organization process itself. Such knowledge is useful, for example, in deciphering the connection between spatial structures and a frequently encountered complex rheology of flowing particulate systems (6), designing adaptive locomotion strategies and decentralized-decision modular robots (7) and swarm robotics (8), and understanding behavioral patterns of active matter (9). For a wide variety of such systems, the basic ingredient of the feedback processes should be, but seldom is, achievable by a set of simple rules via which the individual component responds to a changing environment.

Over the last decade, flow properties of moving granular systems, as archetypes of driven dissipative systems, have been extensively studied (10, 11). The emphasis in these studies has been on understanding their flow behavior (10), occasional loss of translational mobility caused by jamming, which leads to self-organization of stress patterns (12, 13), and differential friction-induced granular segregation (14). The extended nature of individual grains plays a significant role in determining the behavior of the system. For example, a spherical grain driven over a surface may either roll or slide; the two situations involve very different coefficients of friction. Although rolling is known to be an important aspect of granular motion, it has not yet received enough attention (15).

In this paper we present results where a collection of identical spheres moving over a surface forms a self-organized, robust, and protocol-independent steady-state structure. Such a structure is brought about by a mechanism which involves tuning of the time-averaged coefficient of friction of each sphere by toggling between rolling and sliding, along with larger-scale rearrangements caused by a set of recurring local instabilities. Specifically, (i) we uncover the feedback mechanisms by imaging the motion, (ii) translate them into a set of simple local rules that use and illustrate the wide variations in the underlying local dynamics, which give rise to the robust shape, (iii) demonstrate that the key component is a binary alternative between rolling and sliding motion for individual spheres, dictated by collisions among neighbors and, finally, (iv) convert these observed rules into an algorithm that precisely simulates the observed self-organization.

Experimental Details

Fig. 1*A* describes the experimental system under consideration. A moving potential well with a single minimum is generated for an assembly of glass spheres by driving a vertically placed cylindrical glass container of radius R ($=8$ mm) on a circular orbit of radius d ($=2.5$ mm) centered at point C , at a constant frequency f . The geometry of the experimental setup is shown in top view in Fig. 1*B*. In this view the cylinder appears as a circle whereas its axis appears as a single point. We call this point the “center of the cylinder.” Whereas the center O of the cylinder moves on a circular orbit centered at C , the cylinder itself does not rotate about its axis. The yellow shade is the area swept by the cylinder as its axis passing through O moves on a circular orbit of radius d ($=CO$). The line originating from C and passing through O meets the surface of the cylinder at the point M . As O orbits about C (marked by the black dotted line of radius d) the point M moves on a circular path (red dash-dotted line) of radius $(R+d)$ with its center at C . The mechanics of the system is more

Significance

Self-organization is ubiquitous in nature, although a complete understanding of the phenomena in specific cases is rare. Here we elucidate a route to self-organization in a model granular system. The local rules of motion are extracted from the experiment. When converted into an algorithm, they simulate the main aspects of the experimental results. From this, a key ingredient for achieving robustness emerges, namely, a continuously variable relative fraction of time the objects spend in two distinct motional degrees of freedom, rolling and sliding. In so doing, they access a large range of effective friction coefficients that allows self-tuning of the system to adjust its response to changing environments and guarantees a protocol-insensitive unique final state, a previously unidentified paradigm for self-organization.

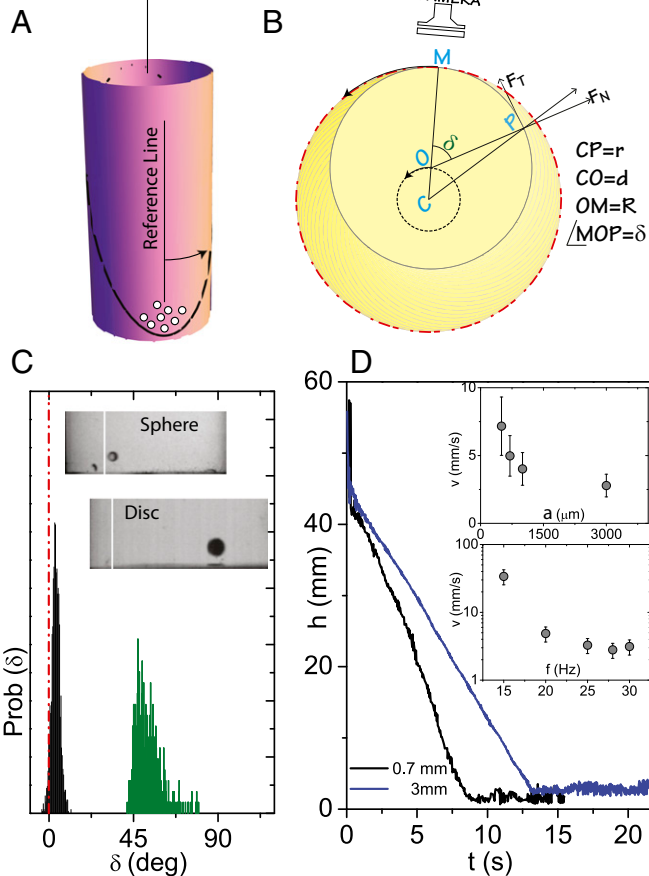
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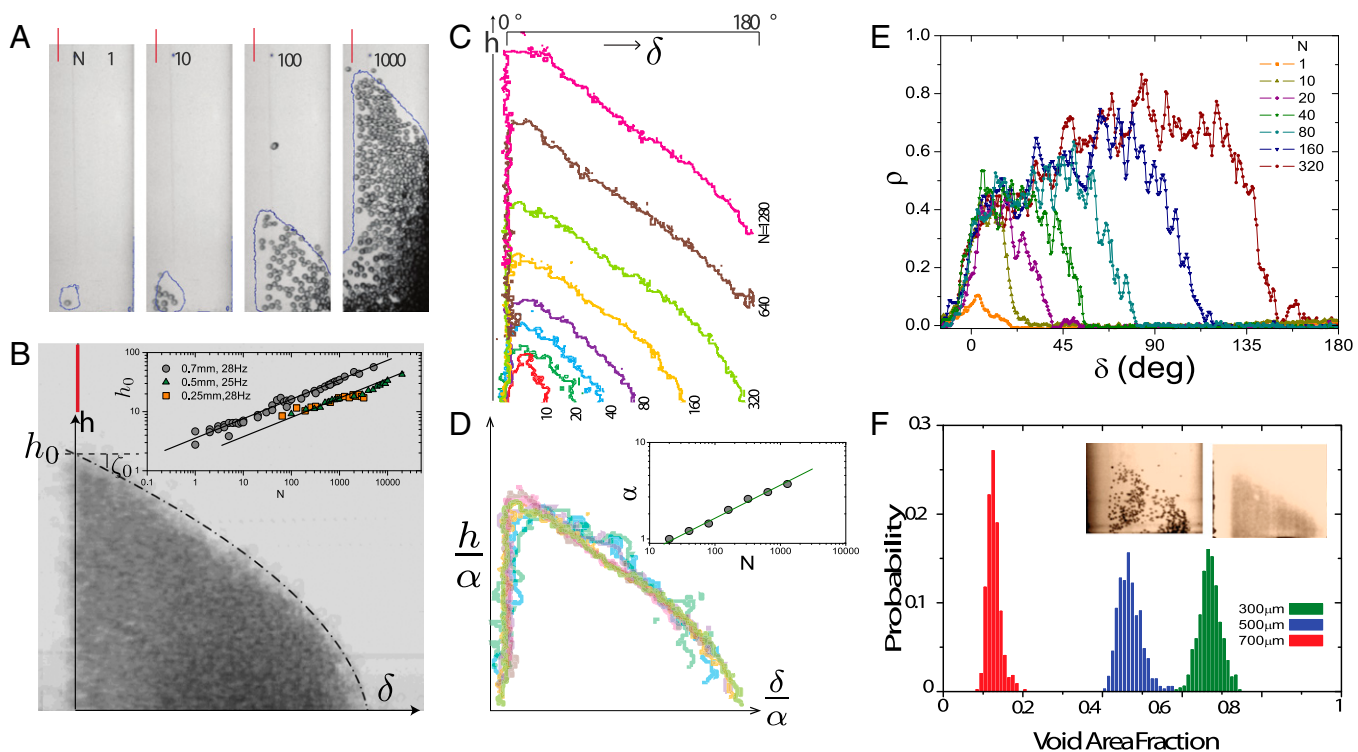


Fig. 2. (A) Images show structures formed at $f = 28$ Hz for $N = 1, 10, 100$ and $1,000$ and $a = 700$ μm . The blue lines mark the outlines of the average of 500 such images. (B) Entire angular profile averaged over 50 cycles of the structure obtained by the stitching procedure described in the text. (Inset) Variation of h_0 with N . The lines marked in the graph correspond to $h_0 \sim N^{0.33}$. (C) Outlines of the shape obtained from images such as in B for various values of N marked besides the curves. (D) The outlines for various values of N as shown in C can be collapsed onto a single curve by scaling both the axes by a factor $\alpha \sim N^{0.33}$ (Inset). (E) Angular variation in density of the structure at a fixed height for various values of N . All of the curves fall on a single master curve. (F) The void fraction in the structure for three different sizes of spheres, $a = 700, 500$, and 300 μm . (Inset) A typical structure for $a = 300$ μm (Left Inset) and the average of many such images (Right Inset).

forces determines the angular position of a single sphere. If we add a second sphere and if the two spheres were to behave like a rigid object with a common coefficient of friction, the force balance requirement would demand an attendant decrease in the angular position of the first sphere. This decrease would scale with the number of spheres in a given row. The observed structure is formed of a number of rows stacked over each other; the length of the rows grows progressively with increasing depth. The occurrence of the vertical front edge that coincides with the reference line suggests the presence of internal dynamics that screens the transmission of stress from the rear to the front of the structure. In this sense it is like a dynamic analog of the Janssen effect observed in static granular matter where stress at the bottom of a granular column saturates exponentially with height (11).

In the imaging scheme described above, as the angular extent of the 2D structure formed on the wall of the cylinder increases, the rear and the front end overlap due to the geometrical constraint, complicating the visualization of the full structure. To overcome this limitation we record a fast movie (at n frames per second) over multiple cycles. From each frame, a strip of angular width $2\pi(f/n)$ is extracted from the central region of the image. Such (n/f) stripes in a cycle are stitched together to obtain the full shape. The stitched images over many cycles are averaged to obtain a time-averaged structure shown in Fig. 2B. Whereas the leading edge is a vertical line, the rear is curved as shown by the dotted line in Fig. 2B.

Fig. 2C shows the outline of the shape for different values of N . We observe that whereas h_0 grows with N as a power law with exponent ~ 0.33 (Fig. 2B, Inset), the angle ζ_0 formed at the vertex by the rear with the horizontal remains constant. The outlines of the shapes corresponding to various values of N collapse onto a single curve by scaling both the δ and h axes by a single parameter, α . The scaled curves are shown in Fig. 2D. The scaling parameter α varies as $N^{0.33}$ (Fig. 2D, Inset), consistent with

$h_0 \sim N^{0.33}$ (Fig. 2B, Inset). In addition, it implies that the angular width of the structure also scales as $N^{0.33}$. The area of the structure grows sublinearly with N .

We note, for example, from the third panel in Fig. 2A that (i) the structure observed is neither close-packed nor of uniform density and (ii) the tendency of the system is to self-organize such that vacancies coalesce to form large voids. The angular dependence of the average number density at a fixed height for increasing number of spheres is shown in Fig. 2E. The structure is denser at the rear than at the front. Unlike the shape which grows in a self-similar manner, the angular profile of density at a fixed height follows a single master curve, independent of N ; the observed departure from the master curve at the rear is related to finite-size effects. The average density also depends on the size of the sphere. The distribution of the void fraction observed in the structure for three different values of a : $700, 500$, and 300 μm for $f = 28$ Hz is shown in Fig. 2F. Smaller spheres are more mobile and hence maintain greater intersphere separation. This is reflected in the variation of density of the structure with the diameter of the sphere. Nevertheless, the average shape remains robust. This is illustrated in Fig. 2F (Insets) for spheres of 300 - μm diameter.

For a constant N , the structure becomes taller with increasing frequency and its width decreases such that $\sin(\zeta_0) = kf^2$ (Fig. 3A) where $k = 0.8 \times 10^{-3} \text{s}^2$ is a constant evaluated from a linear fit. The dependence of height h_0 on frequency is shown in Fig. 3B. The blue curve is the function $h_0 = \sqrt{2Akf/(1 - k^2f^4)}^{1/4}$ obtained, assuming $\sin(\zeta_0) = kf^2$. Here $A \sim (1/2)\delta_w h_0$ is the area occupied by the average structure and δ_w is the width of the structure at zero height. This above functional form fits the data well. At frequencies lower than 15 Hz, the 2D character of the arrangement of spheres on the surface of the cylinder is lost and the system exhibits an entirely different class of phenomena (17).

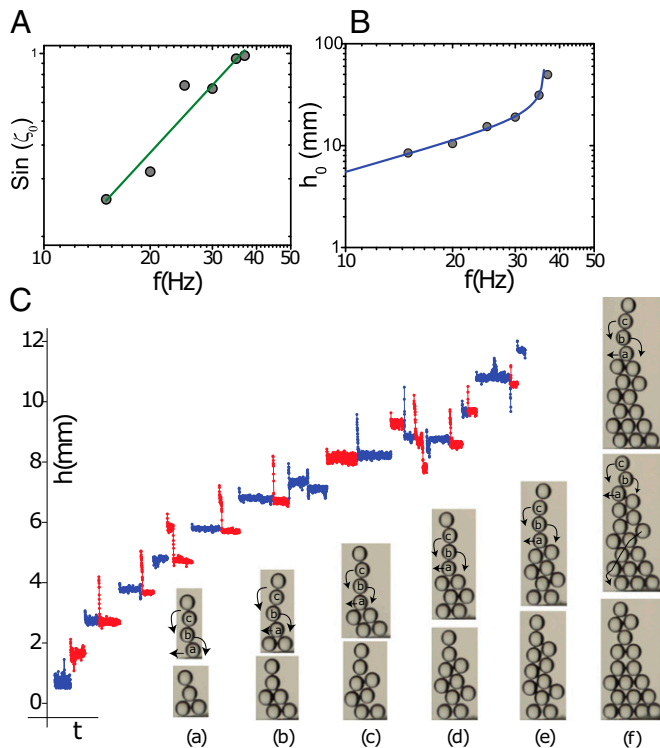


Fig. 3. (A): Variation of $\sin(\zeta_0)$ with f for constant N . The line corresponds to $\sin(\zeta_0) = kf^2$, here $k = 0.8 \times 10^{-3} \text{ s}^2$. (B) Variation of h_0 with f . The blue curve corresponds to the function $h_0 = \sqrt{2Akf/(1 - k^2f^4)}^{1/4}$ with A being the area covered by the structure and is assumed constant. (C) Evolution of the maximum height h_0 of the structure as spheres of size $a = 3 \text{ mm}$ are sequentially added to the system in the presence of the orbital driving. The color flips between red and blue as a new sphere is incorporated into the system. The spheres stack on top of one another. When N attains certain values for example $n = 4, 6, 8, 10, 12, 18$, etc., the system undergoes instabilities that cause rearrangement of the spheres as shown in (C, a–f).

We also explore the dynamics underlying the growth of the structure by sequentially inserting spheres into the system and observing the formation of the structure. To track individual mechanical events in the structure, we choose spheres of diameter 3 mm which have an adequately low mobility (Fig. 1D, *Inset*). Fig. 3C shows the evolution of the maximum height as spheres are added to the system. The color of the plot flips between red and blue as a new sphere gets incorporated in the structure. The first three spheres line up vertically, in a stable configuration. The addition of the fourth sphere leads to a buckling instability toward the right (Fig. 3C, a). This instability lowers the height of the system by the diameter of a single sphere and increases the width by the same amount. This instability occurs periodically (Fig. 3C, b–f) at the top of the structure. For $N \geq 18$ an additional instability which is similar to an edge dislocation (shown by an arrow in Fig. 3C, f, *Middle*) is observed. This experiment shows that mechanical instabilities are intrinsic to the growth and dynamics of the structure.

Prevention of stress transmission in the system is possible when a local force balance is obtained everywhere in the system. The forces in the circumferential direction are the tangential component of the centrifugal force (F_T) and the force of friction (F_F), which depends on the normal load on the sphere and the coefficient of friction. Although the coefficient of friction is a constant between a given pair of materials, for a sphere two distinct values are possible, corresponding to pure rolling and pure sliding, respectively. The two values are well separated (Fig. 1C). In addition, by intermittently rolling and sliding the spheres can have a time-averaged value of coefficient of friction which is

intermediate between these two extreme values. In our system, the force balance between F_T and F_F at the scale of a sphere is achieved through autotuning of μ by intermittent rolling and sliding motion. The spheres close to the leading front are in a purely rolling state whereas those in the rear are sliding. In the intermediate position they intermittently shift between the purely rolling and the sliding states. Hence in the present system the time-averaged coefficient of friction is a self-adjusting quantity whose value depends on the angular position. Similar to sand piles, the slope ($\tan \zeta = dh/d\delta$) of the rear arises from a balance between the force of gravity and friction: $mg \sin(\zeta) = \mu m(2\pi f)^2(R + d \cos(\delta))$. In particular at the leading edge, where $\delta \rightarrow 0^\circ$ and $\zeta \rightarrow \zeta_0$, $\sin(\zeta_0) = \mu(R + d)(2\pi f)^2/g$. This is consistent with the experimental observation in Fig. 3A. In the present system, friction increases with angle and so does the slope $\tan \zeta$ (Fig. 4A).

As a further experimental confirmation of the autotuning of friction, we present results from experiments using donut-shaped objects (shown schematically in Fig. 4B, *Inset*). For these donuts, there are two possible orientations on the wall. In the orientations marked (a) and (b) (Fig. 4B, *Inset*) the donut rolls and slides, respectively, on the underlying surface. The structure in Fig. 4B shows that at the leading front the objects roll [orientation (a) in Fig. 4B], whereas toward the rear they slide [orientation (b) in Fig. 4B]. The existence of a dynamic equilibrium between the rolling and the sliding donuts is shown in Fig. 4B

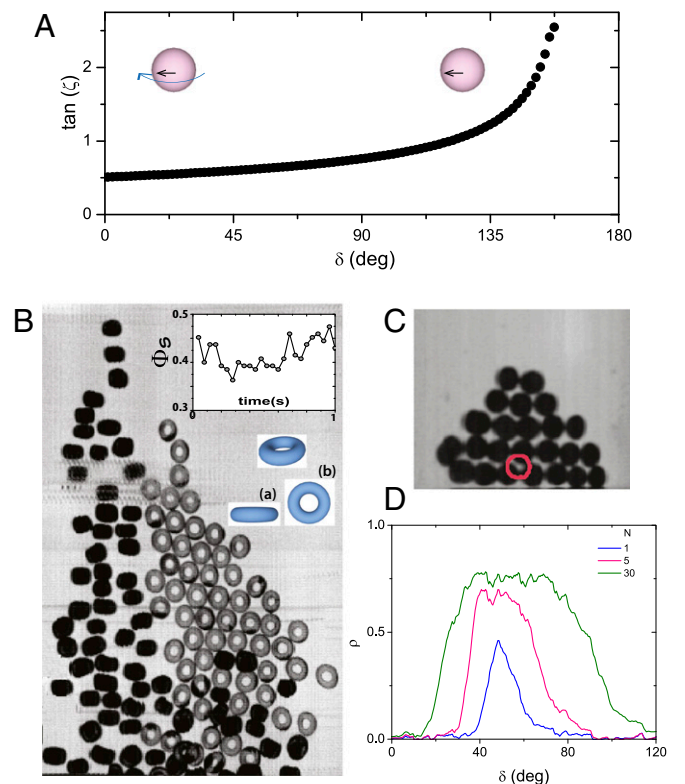


Fig. 4. (A) Variation of the slope, $\tan(\zeta)$ with δ . For small values of δ , $\tan(\zeta)$ is small (~ 0.5) and it corresponds to pure rolling, and for large values of δ , $\tan(\zeta)$ is large (~ 3) and it corresponds to sliding motion. (B) For donut-shaped objects (*Inset*) two possible orientations exist. In orientation (a), the donut rolls on the surface whereas in orientation (b), it can only slide. The rolling donuts are near the leading edge whereas in the rear they are sliding. (Top *Inset*) Fraction of sliding donuts (Φ_s) plotted as a function of time. (C) For disk-shaped objects (diameter 1.2 mm, thickness 0.1 mm) the only possible configuration is to slide. The structure formed for $N = 30$ is symmetric about the red circle which marks the typical position of a single disk. (D) The symmetric growth of angular variation of density with N for the case of disks.

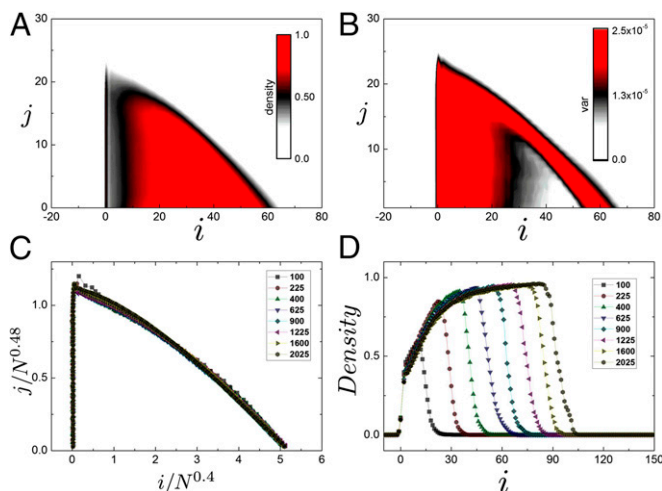


Fig. 5. Results from event-based simulations. (A) Steady-state structure averaged over many iterations for $N = 625$. The color bar represents the site occupation density. (B) A spatial map of the variance $[\text{var}(i,j)]$ of the site occupation measured over many iterations in steady state for $N = 625$. (C) The scaling of the shape with respect to the number of objects. This is obtained by dividing the height index j with $N^{0.48}$ and the angular index i with $N^{0.4}$. (D) The variation of density as a function of index i at a fixed height ($j=6$) follows a single master curve and is independent of N . The symbols and their adjacent numbers shown in C and D correspond to different values of N .

(Top Inset), where the fraction of sliding donuts (Φ_s) and therefore the fraction of sliders too are shown to vary with time.

Moreover, the rolling objects are sparser than the sliding ones. This difference in density arises because of the difference in interobject interaction in the two cases. When objects rolling in the same sense touch each other, the two surfaces in contact move in opposite directions and this hinders the rolling motion (18). As a result, rolling objects tend to stay apart from each other, resulting in a rarer structure. This repulsive interaction is absent for objects that are sliding, which accounts for the variation in density with angular position shown in Fig. 2E. To further test the hypothesis that toggling between rolling and sliding is crucial for the observed phenomenology, we use thin disk-shaped objects which mostly slide. The shape of this object rules out the possibility of autotuning of friction. Fig. 4C shows a typical structure obtained for aluminum disks (diameter 1.2 mm, thickness 0.1 mm). The red circle marks the typical position of a disk in an experiment where $N = 1$. We note that, unlike spheres, the structure for $N = 25$ disks grows symmetrically about the red mark. This is also evident from the symmetrical growth of the angular variation in density for disks and is shown in Fig. 4D for three different values of $N (= 1, 5, 30)$ at a fixed height.

Event-Based Simulation

From the above observations we conclude that with respect to the reference line, an object that can both roll and slide spends most of its time in one of the following states: (i) rolling in place at the reference line, (ii) rolling toward the reference line, (iii) sliding in the positive δ -direction, and (iv) sliding with slipping at a stationary value δ . Only a small fraction of time is spent in changing states and carrying out motions like sedimentation and buckling which are discussed below. Rolling is possible for horizontally isolated objects. For objects that touch each other in the horizontal direction, rolling produces shearing of the contact zone and hence is an unfavorable mode of motion of the object (18). Blocks of horizontally connected objects thus tend to slide. Thus, the transition from the state of rolling to the state of sliding is triggered by the formation of contacts between a rolling object and its neighboring objects; these contacts are broken stochastically by

mechanical noise which occasions the opposite kind of transition, from the sliding state to the rolling state. These recurrent local transitions between rolling and sliding of the objects bring about and maintain an overall balance of forces within the structure. These transitions result in the formation of regions of higher dissipation of energy (sliding) and lower dissipation of energy (rolling) at the rear and front of the structure, respectively. The dissipated energy is continuously replenished from the walls of the driven cylinder. Our observation of the local behavior of the objects is translated into the following empirical rules:

C₁ (Sedimentation) An object moves downward if there is room below it.

C₂ (Rolling) A horizontally isolated object away from the reference line rolls toward the reference line.

C₃ (Sliding) A horizontally connected block of objects moves together in the positive δ -direction. This motion persists until the last object (farthest from the reference line) in the block reaches a maximum value of δ determined by the sliding friction coefficient, after which this block slides in place.

C₄ (Buckling of columns) Vertical columns of connected objects buckle.

C₅ (Buckling of rows) Horizontal columns of connected objects buckle.

C₆ (Breakaway) Mechanical noise stochastically breaks the first object (closest to the reference line) away from a block of sliding objects.

Starting from the empirical rules we construct a set of algorithmic rules for the time evolution of the system of such objects. Based on these rules, an event-based computer simulation is carried out on a 2D rectangular array of dimension 201×200 . The entries in the array are indexed by pairs of integers: (i,j) , where $-100 \leq i \leq 100$ and $0 \leq j \leq 199$. The coordinate i increases linearly with δ and j denotes the height. The column $j=0$ is the reference line and $i_{\text{max}}=80$ is the maximum distance that a connected block of objects can slide to. Physically it is related to the coefficient of sliding friction. At any instant t of time, a location (i,j) may either be occupied or unoccupied. We describe the state at any instant t by a function $f_t(i,j)$, where $f_t(i,j) = \begin{cases} 1, & \text{if the } (i,j)\text{th site is occupied} \\ 0, & \text{if the } (i,j)\text{th site is vacant} \end{cases}$. The algorithmic rules of evolution treat t as a discrete variable taking integer values and produce the output function f_{t+1} from a given input function, f_t . At each instant of time, we randomly apply one of the six rules $C_1 \dots C_6$. The random choice at an instant t will be denoted by $C(t)$. For this choice, the rule C_i is assigned a probability, p_i . The simulation is done with values $p_1=1/5$, $p_2=1/5$, $p_3=2/15$, $p_4=1/5$, $p_5=1/5$, and $p_6=1/15$. However, nearby values of probabilities give qualitatively similar results. Given a function $f_t(i,j)$ that takes values 0 or 1, a rule chosen from among the six permitted rules $C_1, \dots, C_6$ is applied to the matrix $f_t(i,j)$ in a raster scan manner, starting from the left top corner $(-100,200)$ and continuing row by row until we arrive at the right bottom corner $(100,0)$. This produces the new state matrix $f_{t+1}(i,j)$. The choice of the scanning direction is motivated by the experimental observations (Fig. 3C). The entire process is now iterated, beginning with the updated state $f_{t+1}(i,j)$. This results in the updated occupancy function $f_{t+2}(i,j)$ and so on. The coordinate description of each rule is given in detail in *Methods*.

Results from Event-Based Simulation. The steady-state structure averaged over many iterations from the simulation is shown in Fig. 5A for $N = 625$. The overall features, i.e., (i) a sharply vertical front, (ii) a curved rear, and (iii) a density gradient from the front to the rear, the latter being denser, are in agreement with experimental findings. In the simulations, the spatial variation in intermittency, i.e., the toggling between the objects rolling and

